BTL1213 Operating Systems / BIT1213 Operating Systems / BCL1213 Operating Systems

SEGi University

Chapter 03 Memory Management - Part B



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

BACHELOR OF COMPUTER SCIENCE (HONS)

By: RUFFINUS RUSELL

Learning Objectives:

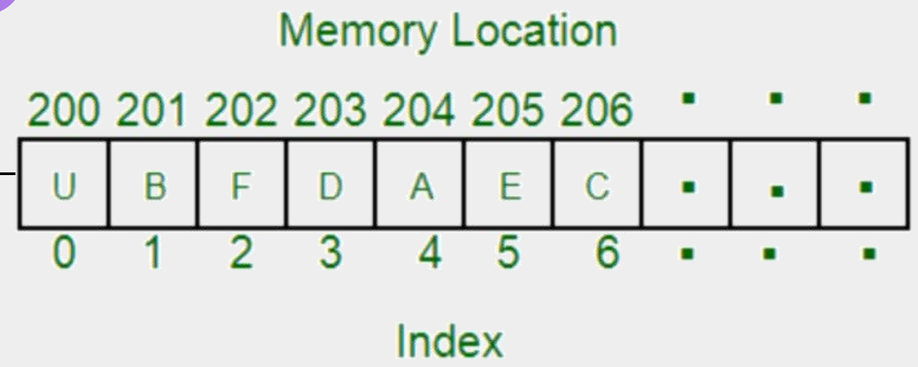
By the end of this lesson, students will be able to:

- Understanding concept of memory and it’s important role.
- Identify different types of memory.
- Analyze examples of memory allocation.
- Explain the factors affecting memory speed.

OVERVIEW

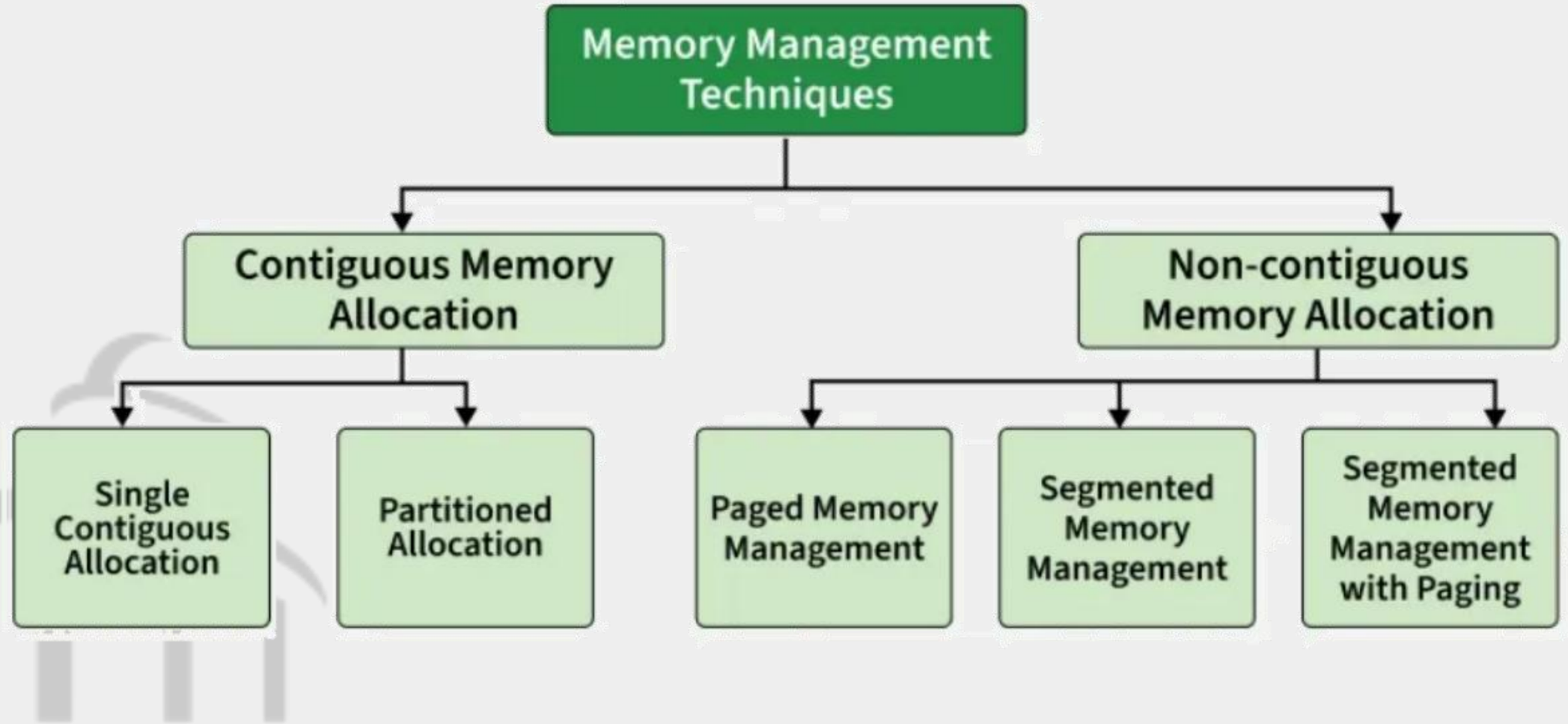
Topic for today's main discussion:

- Definition of Memory
- What is Memory Management ?
- Memory Hierarchy
- Types of Memory Allocation
- Memory Access in Operating System



04 Memory Allocation

- Memory allocation refers to the process of assigning memory space to programs and processes.
- OS uses this meachanism to allow multiprograms run simultaneously.
- It ensures efficient use of available memory while preventing conflicts between different processes.

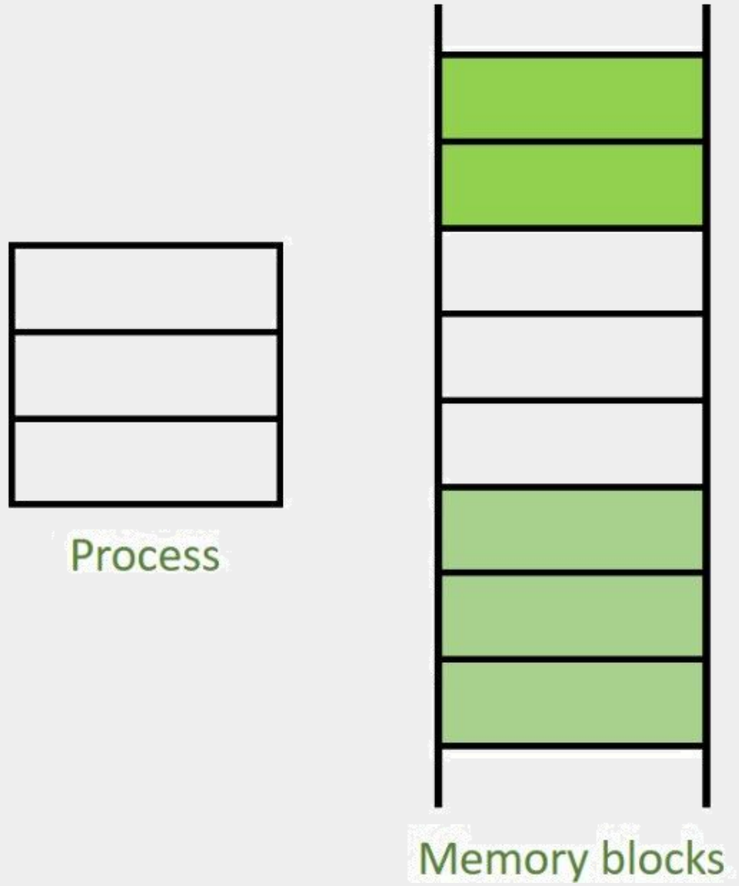

Memory Management Techniques
Contiguous Memory Allocation
Non-contiguous Memory Allocation
Single Contiguous Allocation
Partitioned Allocation
Paged Memory Management
Segmented Memory Management
Segmented Memory Management with Paging

a) Contiguous Memory Allocation

- Contiguous memory allocation is a memory management method where each process is given a single, continuous block of memory.
- This means all the data for a process is stored in adjacent memory locations.

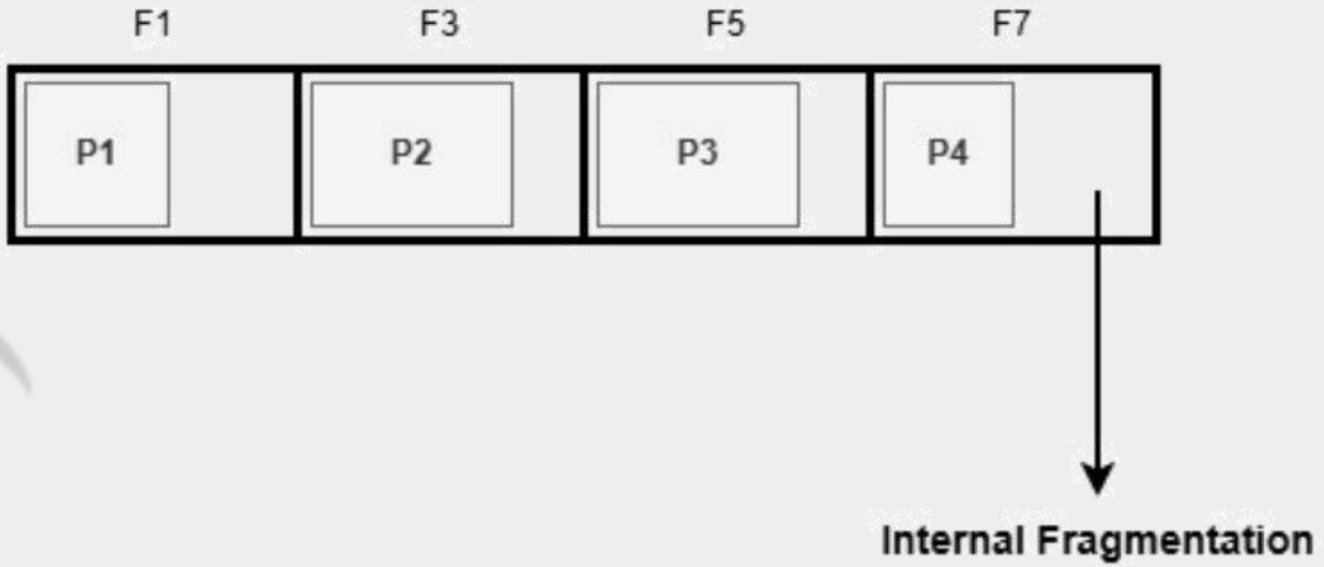
  i) Fixed Partitioning (Static)

  ii) Dynamic Partitioning (Variable)



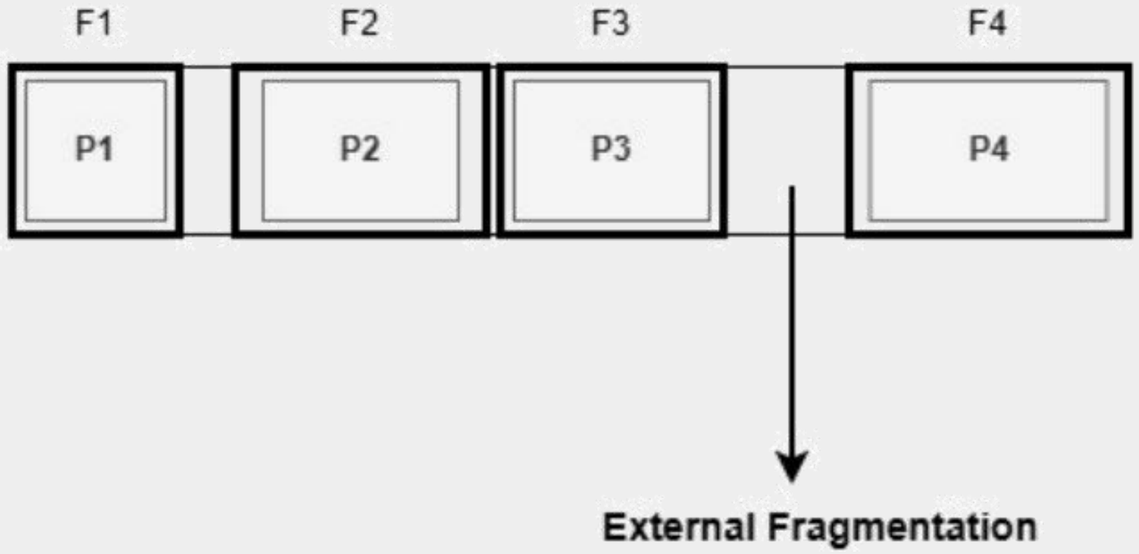
i) Fixed Partitioning (Static)

- Main memory is divided into partitions of fixed sizes.
- Each partition can hold exactly one process. When a new process loaded into memory, it is allocated a partition of fixed size.



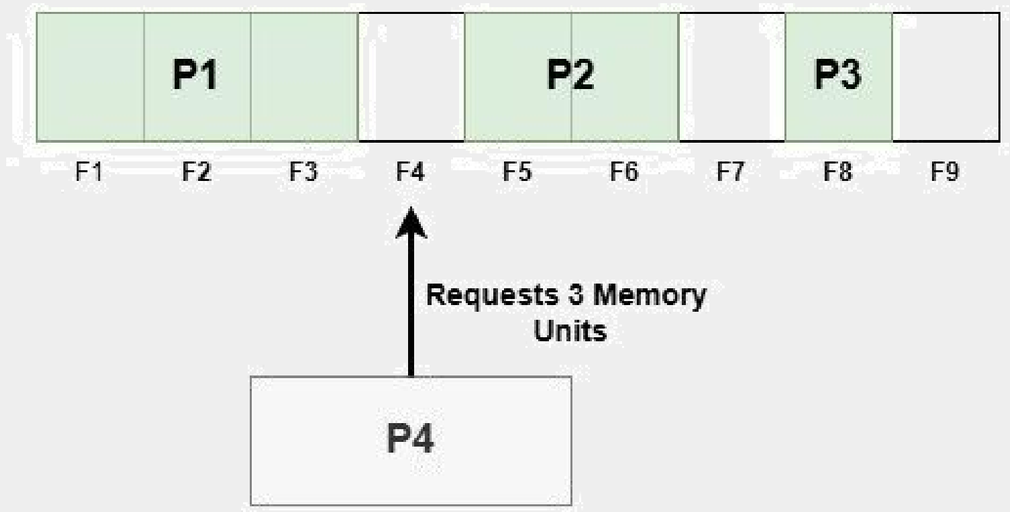
ii) Dynamic Partitioning or Variable Partioning

- Main memory is divided into partitions of varying sizes depending on the size of the processes being loaded into memory.
- When a new process is loaded, it is allocated to the first available partition that is large enough to hold the process.



Why Fragementation?

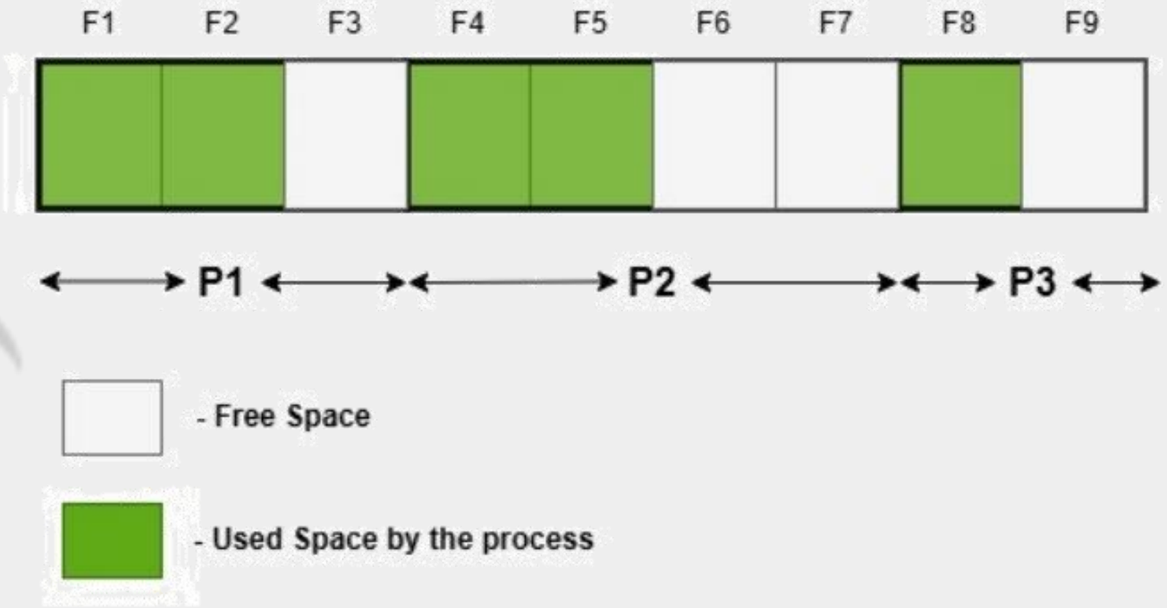
- Fragmentation is a condition where the available memory is divided into small and non-contiguous blocks after allocating and deallocating memory to various processes.
- Fragmentation = gap.
- Fragmentation can lead to wastage of memory.



Types of Fragementation

i) Internal Fragmentation

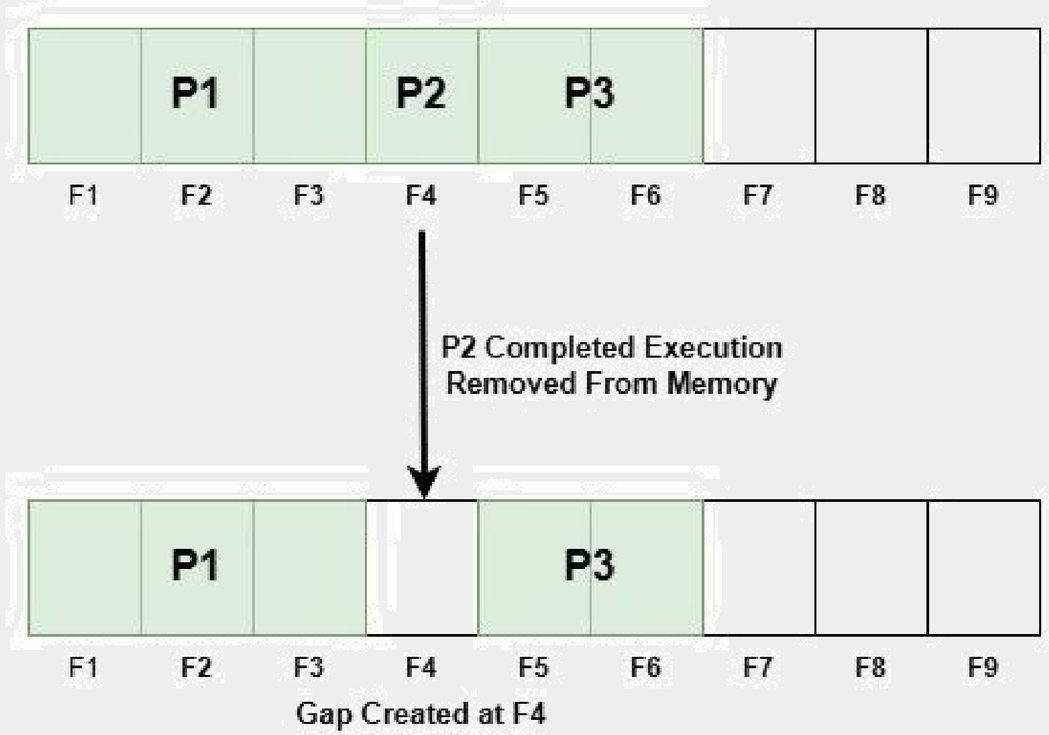
- Occurs when a process gets more memory than it actually needed.
- This can happen for large number of processes running in the system, which will lead to wastage of large amount of memory.



- Causes of Internal Fragmentation:
    a) Fixed Size Memory Allocation
    b) Management Overheads
    c) Memory Alignment

ii) External Fragmentation

- Refers to gaps generated in memory when multiple processes are loaded and unloaded from memory.

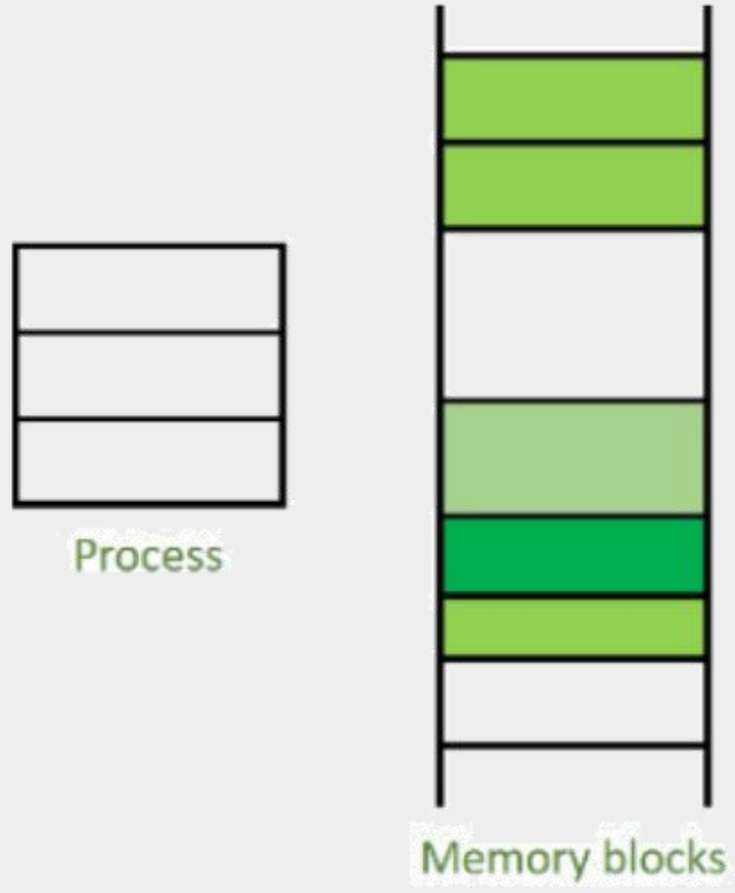


- Causes of Internal Fragmentation:
    a) Dynamic Memory Allocation
    b) Process Termination
    c) Variable Siza Memory Allocation

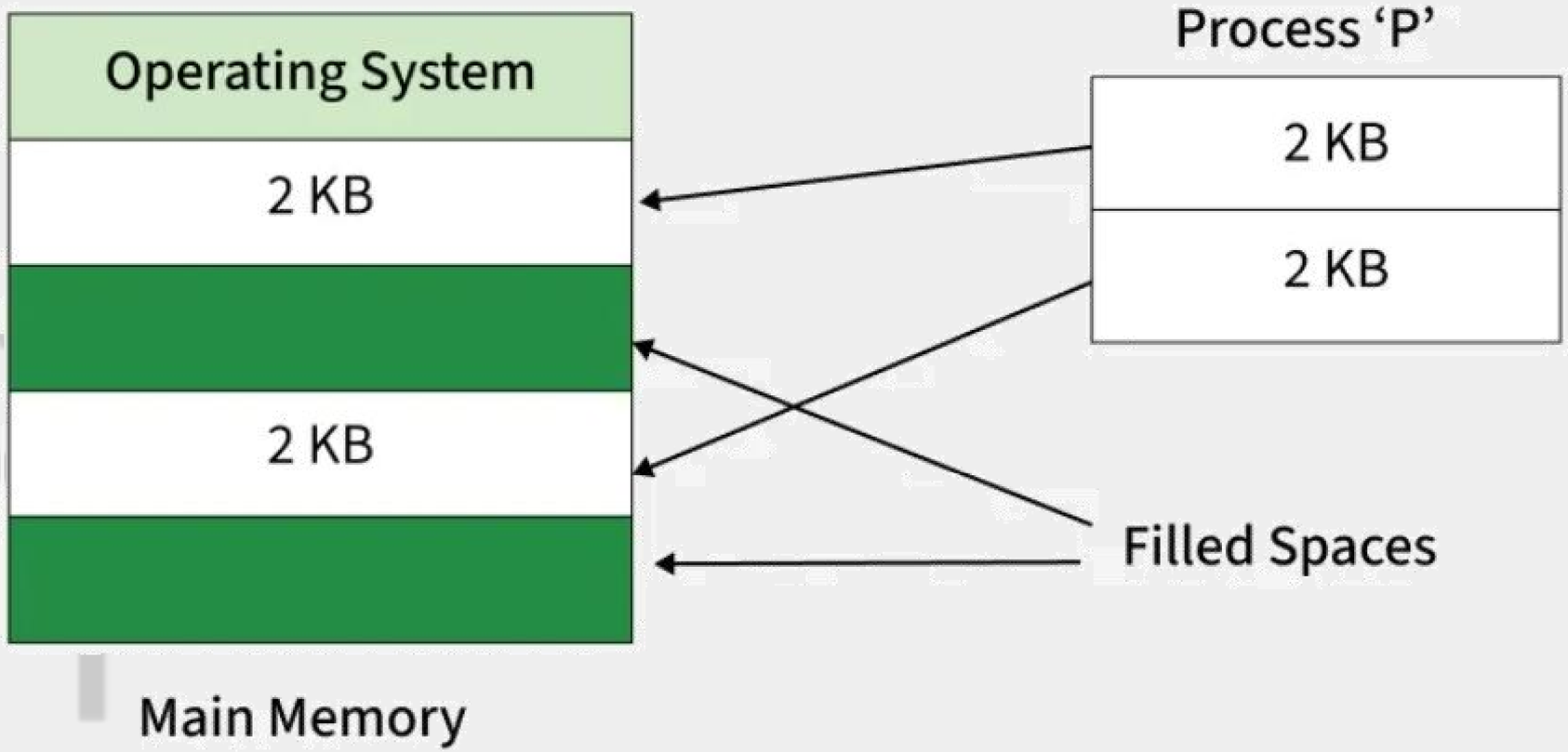
- How to reduce fragmentation?
    a) Compaction
    b) Paging
    c) Segmentation

b) Non-Contiguous Memory Allocation

- Technique to allocate memory to processes that do not require a contiguous block of memory.
  - Process is split into smaller units (pages/segments).
  - These parts are placed in available free blocks (frames) of memory.
  - OS maintains mapping through page tables or segment tables.

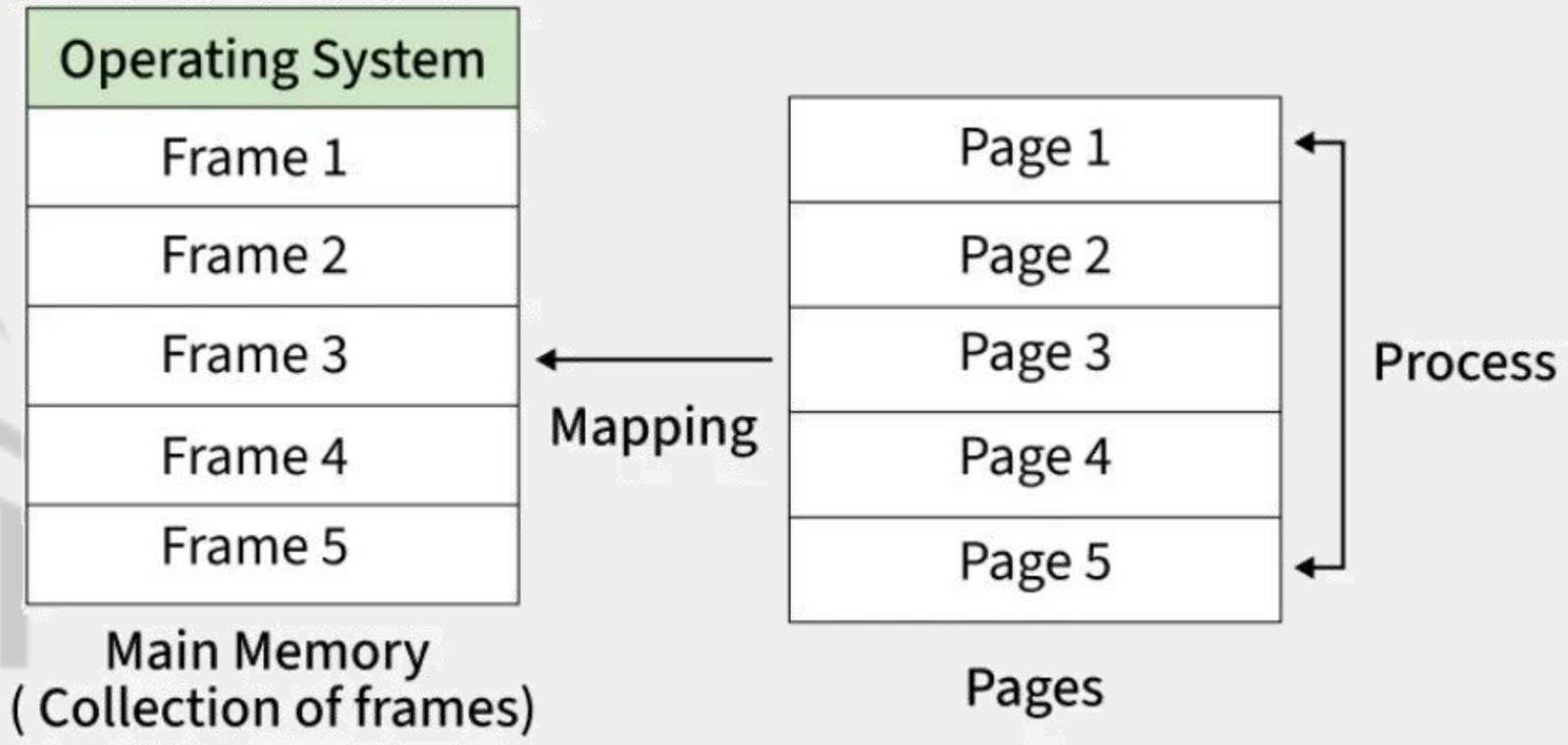


• The process can be divided into different parts hence filling the space in the main memory.



i) Paging

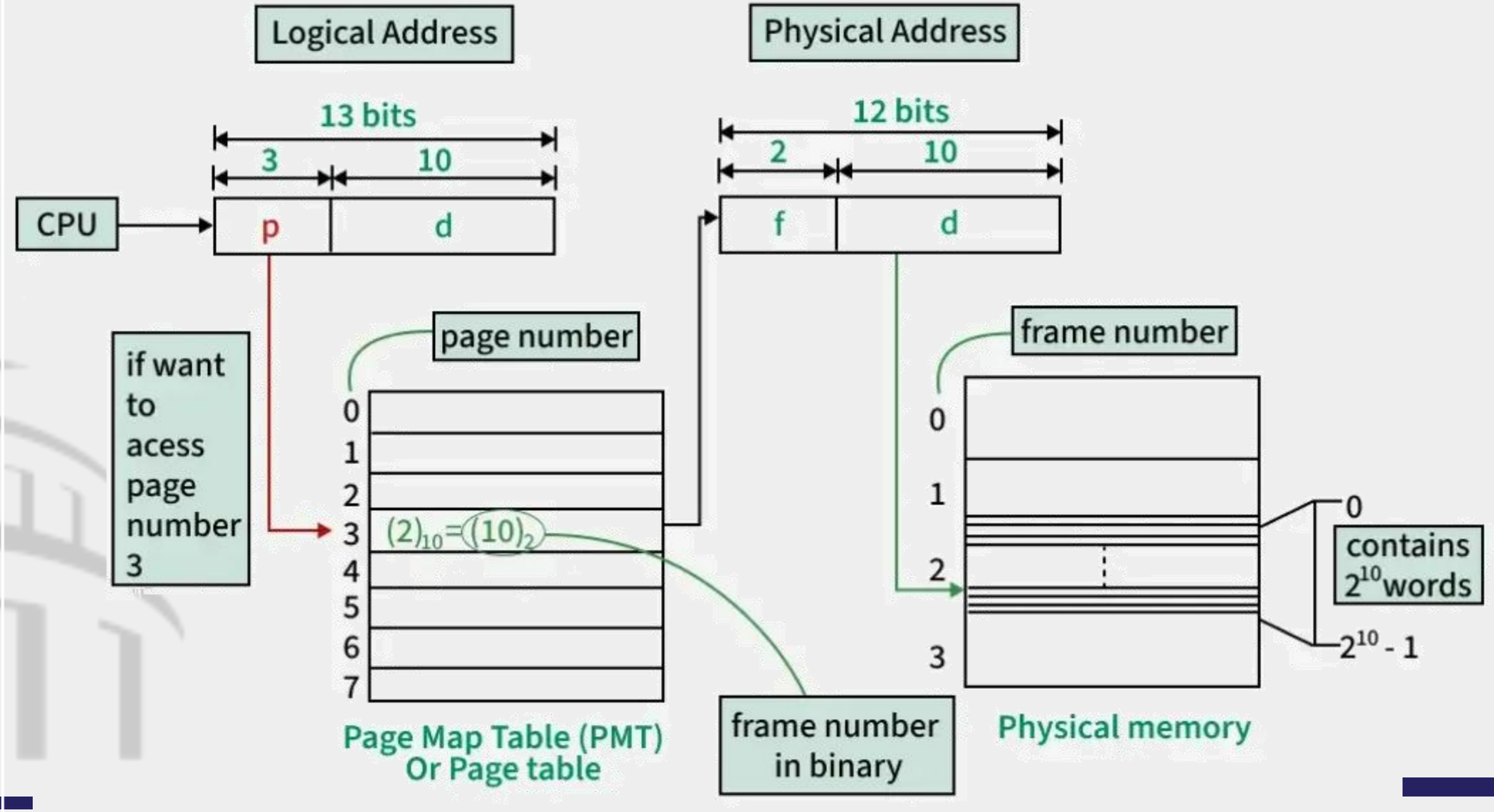
- A process is divided into fixed-size blocks called pages, and physical memory is divided into frames of the same size.



i) Paging in Memory Management

- Memory isn’t always available in a single block.
- Processes size can increase or decrease.

ii) Working of Paging



iii) Memory Management Unit (MMU)

- The Logical Address Space is also split into fixed-size blocks, called pages.
- The Physical Address Space is conceptually divided into a number of fixed-size blocks, called frames.
- Page Size = Frame Size

number of pages = Logical Address Space / Page Size

number of frames = Physical Address Space / Page Size

a) Example 1

- Logical Memory = 16KB
- Physical Memoy = 8KB
- Page Size = 2KB

a) Calculate Number of Pages:



b) Calculate Number of Frames:



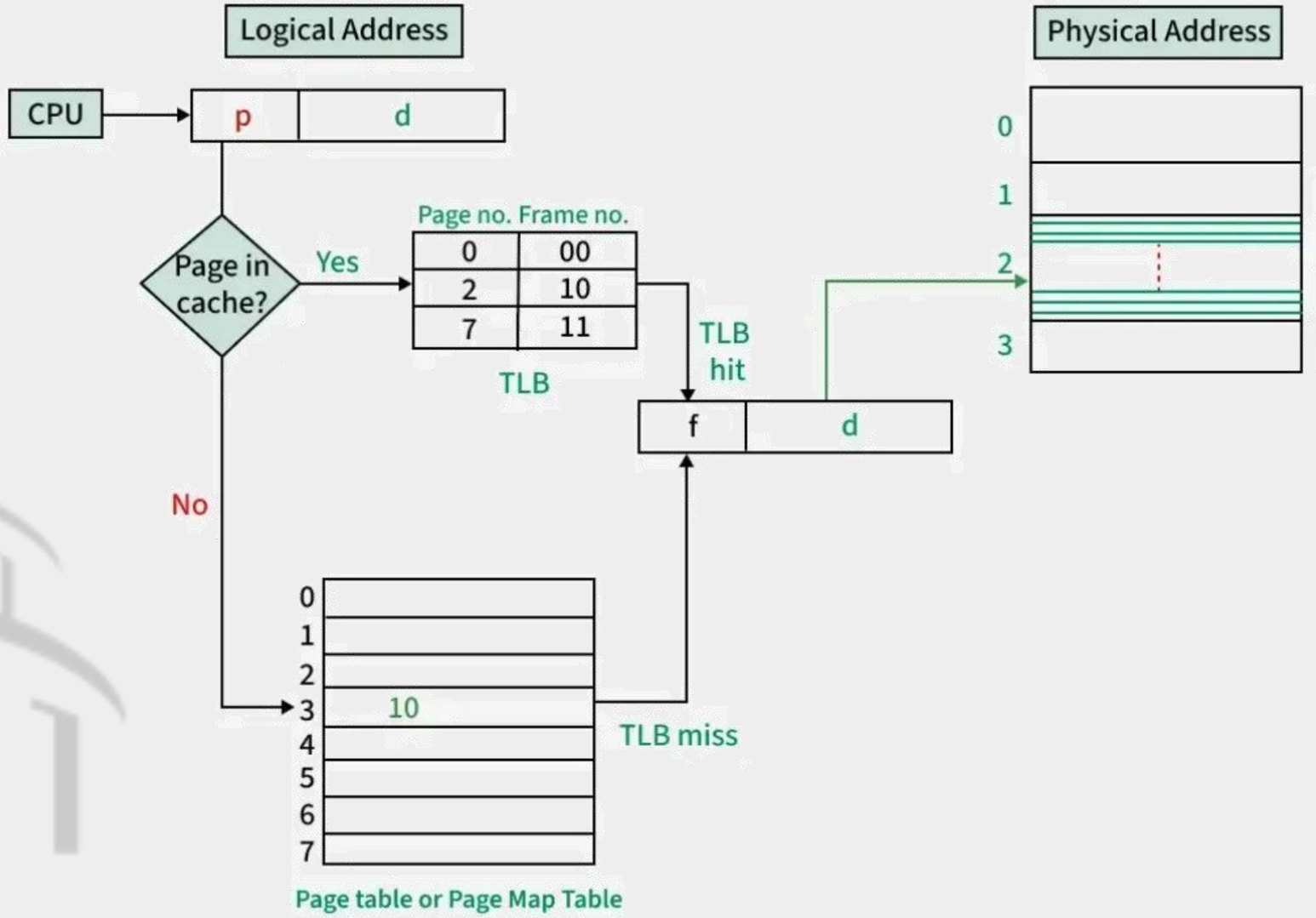
b) Example 2

- Logical Memory = 24KB
- Physical Memoy = 12KB
- Page Size = 4KB

a) Calculate Number of Pages:

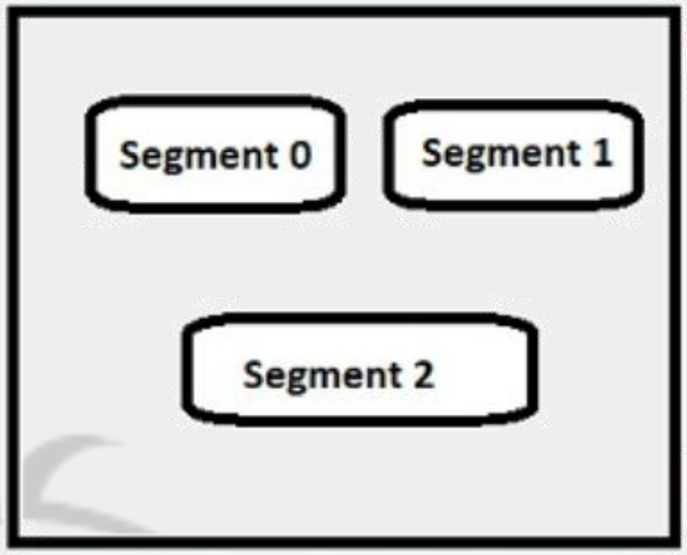
b) Calculate Number of Frames:

iv) Hardware Implementation of Paging



ii) Segmentation

- Technique divides a program's memory into different segments based on the logical divisions of the program.
- Each segment is used to store a specific part of the program, such as functions, code, or variables.
- Segment represents a logical part of a program such as functions, arrays, or modules.



Logical Address Space

	base address	Unit
0	500	600
1	2500	800
2	1500	400

Segmentation Table



Physical Address Space

Base Address = Starting address in physical memory

Unit / Limit = Size of the segment (maximum value offset)

a) Example 1 = Segment 0

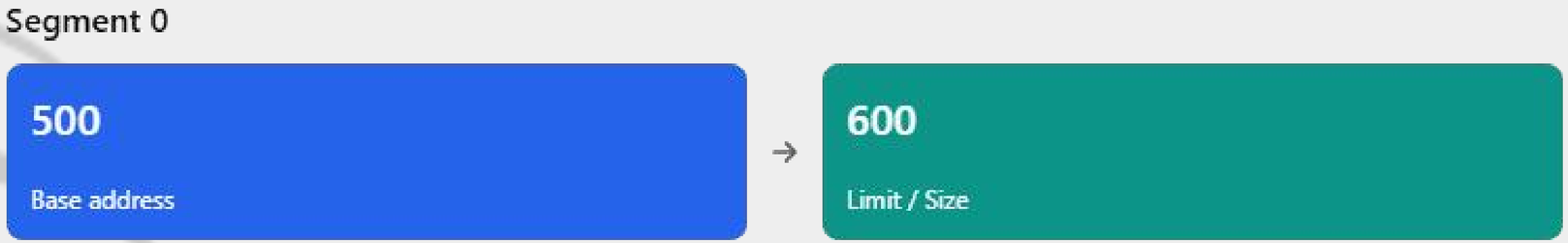
A process wants to access Segment 0 at offset 200.

- Physical Address = Base Address + Offset

  = 500 + 200

  = 700

- The logical address (500, 700) is mapping to segment 0 in physical memory.



*Offset < Limit

b) Example 2

Segment	Base Address	Limit
0	1000	400
1	2500	600
2	4000	300

*Given Offset = 200

Questions:

a. A process wants to access Segment 0 at offset 200. Calculate the physical address.

b. A process wants to access Segment 1 at offset 500. Determine whether the logical address is valid. If valid, calculate the physical address.

c. A process wants to access Segment 2 at offset 350. Determine whether the logical address is valid. Explain your answer.

d. A process wants to access Segment 1 at offset 600. Determine whether the logical address is valid. Explain your answer.

e. A process wants to access Segment 2 at offset 100. Calculate the physical address.

Why Segmentation?

- Reduced Internal Fragmentation
- Smaller Segment Table
- Better CPU Utilization
- Closer to User’s View
- User-Controlled Size
- Security & Separation

Paging Vs Segmentation

Paging

- Invisible from the user program
- Divides physical memory into blocks (pages)
- Pages are of uniform, predetermined size
- Logical address is a positive integer

Segmentation

- Visible to the user program
- Divides logical memory into blocks (segments)
- Segments are of arbitrary sizes
- Logical address is an ordered pair: (segment, offset)

iii) Segmentation with Paging

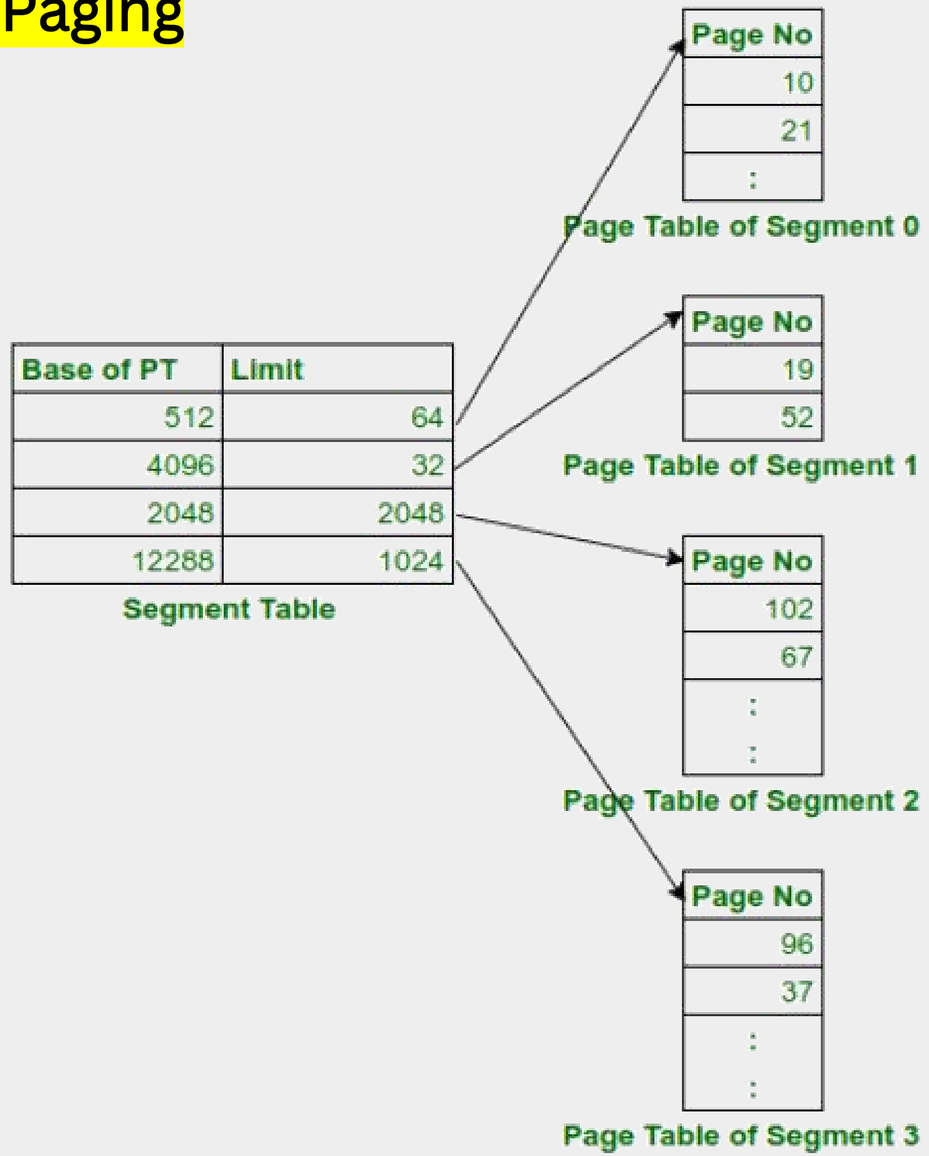
- Thus technique combines both segmentation and paging.
- A program is first divided into logical segments, and each segment is further divided into fixed-size pages.
- This approach maintains the logical view of segmentation while reducing external fragmentation using paging.

Working of Segmentation

- A process is generally divided into four segments, namely - code, data, stack and heap.



- View of Segmented Paging



CONTIGUOUS MEMORY ALLOCATION	NONCONTIGUOUS MEMORY ALLOCATION
A memory allocation model that assigns memory blocks having consecutive addresses to a process	A memory allocation model that assigns a process to memory blocks located in different location
Assigns consecutive blocks of memory to a process	Assigns a process to memory blocks located in different location. It follows a nonconsecutive manner
Overhead is minimum - not much address translations while executing a process	More overheads as there are more address translations.
A process executes faster because the whole process is in a sequential block	A process requires more time to execute as it is in different locations of the memory
Easier for the operating system to control	Difficult to control

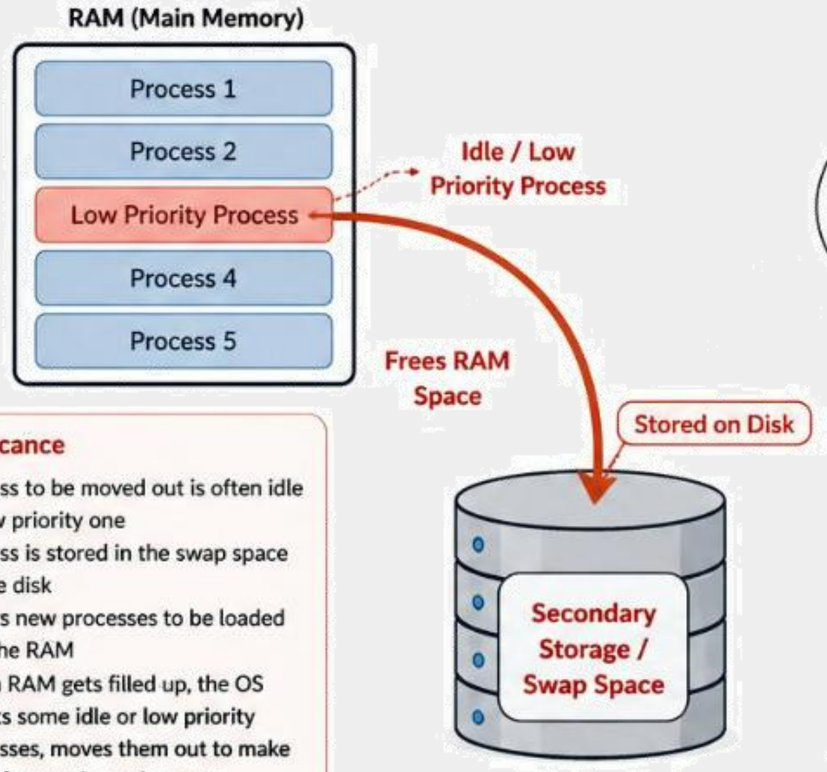
Visit www.PEDIAA.com

Swapping in Memory Management

- Swapping = A process-based is temporarily moved from main memory (RAM) to secondary storage (disk) and vice versa.
- Done through the medium term scheduler.
- This allows the operating system to manage limited RAM effectively and run multiple processes concurrently in a multiprogramming environment.

1. SWAP OUT (ROLL OUT)

Moving the process out from the RAM to the secondary storage.

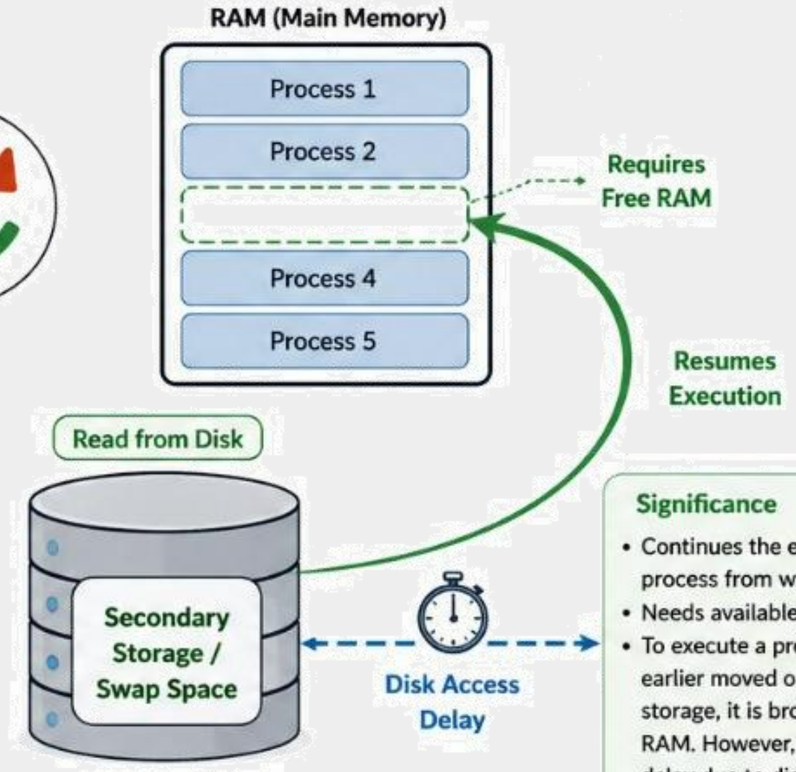


Significance

- Process to be moved out is often idle or low priority one
- Process is stored in the swap space on the disk
- Allows new processes to be loaded into the RAM
- When RAM gets filled up, the OS selects some idle or low priority processes, moves them out to make room for new incoming ones.

2. SWAP IN (ROLL IN)

Returning the process from the disk to the RAM.



Significance

- Continues the execution of the process from where it left off
- Needs available space in the RAM
- To execute a process that was earlier moved out to the secondary storage, it is brought back into RAM. However, there is some delay due to disk access.

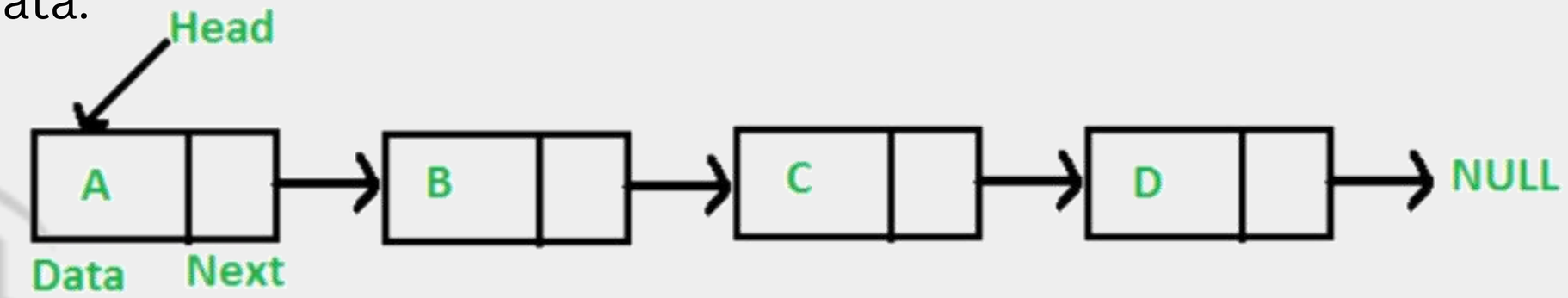
SUMMARY: Swapping helps the OS manage limited RAM efficiently by temporarily moving processes to disk and bringing them back as needed.

05 Memory Access in OS

- Memory access in operating system refers to how a process or program interacts with system memory in an operating system.
- The OS manages memory access to ensure efficient, secure, and error-free operations.
- 4 types of Memory Access methods :
    a) Sequential Access
    b) Random Access
    c) Direct Access
    d) Associative Access

a) Sequential Access

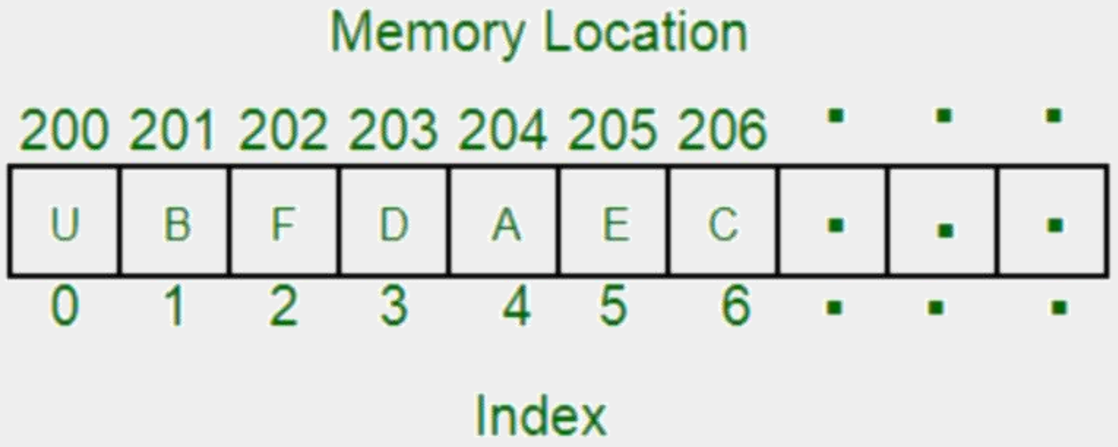
- The memory is accessed in a specific linear sequential manner, like accessing in a single Linked List. The access time depends on the location of the data.



- Applications of this sequential memory access are magnetic tapes, magnetic disk and optical memories.

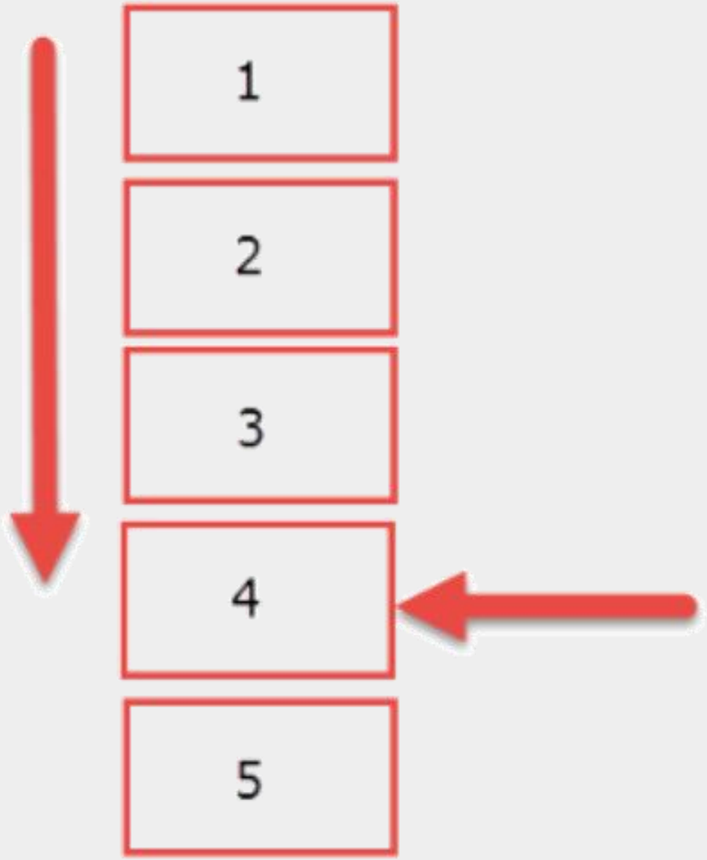
b) Direct Access

- Access is accomplished by direct access to reach a general vicinity plus sequential searching, counting or waiting to reach the final destination.



- Application of thus direct memory access is magnetic hard disk, read/write header.

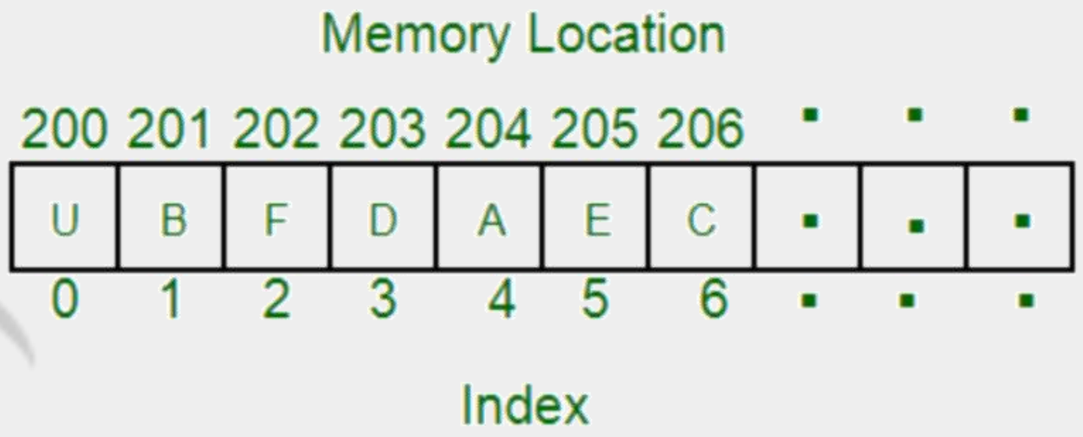
With **sequential access,** elements #1,2, 3 must be processed before element #4 can be processed.



With **direct access**, element #4 in the list can be accessed without having to process the elements before it.

c) Random Access

- In this method, any location of the memory can be accessed randomly like accessing in Array.
- Physical locations are independent in this access method.



- Applications of this random memory access are RAM and ROM.

d) Associative Access

- In this memory, a data is accessed rather than its address.
- Data is located by a comparison with contents of a portion of the store.
- Access time is independent of location or pervious access.
- Application of thus Associate memory access is Cache memory.

Summary

Why we learn Memory Management?

- Memory management is a crucial function of an operating system.
- To ensure protection of different processes from each other.
- To place the programs in memory.
- It comprosises of memory hierarchy that handles memory allocation, tracking, memory access method and optimization to ensure efficient execution of processes in computing system.

THANK YOU



